The Finite-Harmonic Formal Second-Order Tangent Cone of Compact Weyl–Maxwell Gravity

Moment Maps, Exponential-Polynomial Corrections, and Bounded Resonances

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Abstract

Linearized solutions of a nonlinear gauge theory need not be tangent to exact solutions. We classify this failure at second order for finite harmonic configurations of Weyl–Maxwell gravity on the compact magnetically supported product $\mathbb{R}_t \times S^1 \times S^2$. The certified linear inventory contains the ordinary Einstein–Maxwell branches, two additional fourth-order branches in each generic parity, exceptional dipoles, a homogeneous generalized block, and axial twist modes.

For spatially periodic corrections with finite exponential-polynomial time dependence, the complete second-order obstruction space is five-dimensional. Its covectors are precisely the Taub pairings with time translation, circle translation, and the three sphere rotations. Thus the complete finite-support exponential-polynomial tangent cone is

$$\mathcal{Z}_2^{\mathrm{ep}} = \{u : \mu_H(u) = \mu_{P_x}(u) = \mu_{J_1}(u) = \mu_{J_2}(u) = \mu_{J_3}(u) = 0\}.$$

This is a necessary-and-sufficient theorem in the declared correction class, not merely a list of examples.

Bounded corrections obey additional conditions. Polynomial-growth source coefficients and nonzero-shell adjoint pairings survive independently of the five moment maps. We classify the complete standard generalized-zero subcone, prove that electric-charge and flat-Wilson oscillator columns are bounded-removable, and determine the full circumference column: a nonzero circle-radius tangent is compatible with a finite oscillator precisely when that oscillator has zero circle momentum. A repaired full-time circumference-velocity fixture also exposes a nonzero polar polynomial obstruction that was invisible at $t = 0$; its full a, d-cross ideal on the $\ell = 2, k = 0$ extra block is now classified. On the complete global-plus-$\ell = 2, k = 0$-extra carrier, the bounded cone then collapses exactly to the standard static-global cone: no nonzero extra amplitude survives. At one tuned nonzero momentum, by contrast, we classify the complete axisymmetric all-primary bounded cone. Its only shell resonance cuts out two mixed axial–polar Einstein-minus sheets, and the lower-frequency extra primary widens the exact interval on which positive-primary occupations can balance their energy and momentum. The remaining oscillator columns are stated as open gates. A successor theorem promotes the standard-branch part of this calculation to every $\ell \geq 2$: the exact two-parity resonance matrix has two mixed sheets and two one-sided planes, and charge balance reduces the complete tuned axisymmetric bounded cone to the origin together with the two mixed sheets on an explicit occupation interval. A separate two-absolute-momentum computation has certified all 108 axisymmetric $L = 4$ branch-basis coefficients and all 56 declared nonaxisymmetric $L = 1, 3$ coefficients; every individual basis

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fixture is obstructed, while amplitude-level cancellations remain open. Consequently formal exponential-polynomial extendibility is a moment-map theorem, whereas bounded quasiperiodic extendibility is a strict resonance problem.

All results are LOCAL-ALGEBRAIC and REDUCED-MODE. They do not establish an all-orders family, a causal retarded correction, a final residual quotient, or a quantum state space.

AI authorship and computational accountability

The named AI systems developed the programme, derivations, symbolic implementations, independent verification architecture, claim audits, and this manuscript. Asger Alstrup Palm commissioned and coordinated the work and is the corresponding human contact. Every theorem stated as certified below is bound to an exact machine-readable certificate and a separately implemented verifier. Exact rational or algebraic arithmetic is used for all ranks, cokernels, polynomial reductions, and source remainders. The manuscript remains conditional on final human review of its mathematical scope, references, and exposition.

Contents

1 Question, answer, and claim boundary

Let $\Phi(s)$ be a putative curve of solutions of the Weyl–Maxwell equations $\mathcal{E}(\Phi) = 0$, expanded as

$$\Phi(s) = \bar{\Phi} + su + s^2 v + O(s^3).$$

The first two equations are

$$\mathcal{L}u = 0, \qquad \mathcal{L}v = -\mathcal{E}^{(2)}_{\bar{\Phi}}(u,u), \qquad \mathcal{L} := D\mathcal{E}_{\bar{\Phi}}, \qquad \mathcal{E}^{(2)}_{\bar{\Phi}} := \frac{1}{2}D^2\mathcal{E}_{\bar{\Phi}}. \tag{1}$$

The first equation describes the linear solution space. The second asks whether the quadratic source lies in the image of the linearized operator in a specified function class. Changing that function class changes the answer: a resonant source may have a finite exponential-polynomial secular primitive but no bounded quasiperiodic primitive.

The paper answers the following finite-harmonic question.

> For the complete certified linear inventory on the compact product, which real finite-support tangents admit a spatially periodic second-order correction, and which additional conditions appear when that correction is required to remain bounded in time?

The answer has two layers:

$$u \in \mathcal{Z}_2^{\mathrm{ep}} \iff \mu_H = \mu_{P_x} = \mu_{J_1} = \mu_{J_2} = \mu_{J_3} = 0, \tag{2}$$

$$u \in \mathcal{Z}_2^{\mathrm{qp}} \iff u \in \mathcal{Z}_2^{\mathrm{ep}}, \quad \mathcal{P}_{j,r}(u) = 0, \quad \mathcal{R}_{j,a}(u) = 0. \tag{3}$$

Here $\mathcal{P}_{j,r}$ are positive polynomial-growth coefficients and $\mathcal{R}_{j,a}$ are reduced adjoint pairings on nonzero characteristic shells. Both obstruction ledgers are complete in the declared finite carrier. The paper gives several complete columns of the second line but does not claim a classification of the common quadratic zero locus of all bounded obstructions.

1.1 Dependency typing

The spacetime is Lorentzian, but the proof is an exact harmonic and finite-module calculation. It carries the dependency tags

LOCAL-ALGEBRAIC REDUCED-MODE.

No theorem below carries LORENTZIAN-CAUSAL. In particular, a finite exponential-polynomial right inverse is not a retarded Green operator, and a bounded quasiperiodic correction is not an all-orders nonlinear solution.

2 Relation to classical linearization stability

The structural origin of the five quadratic conditions is classical. Brill and Deser exhibited nonintegrable linearized solutions on closed spatial sections [1]; Fischer and Marsden formulated linearization stability as a nonlinear functional-analytic problem [2]; and Moncrief identified spacetime symmetries with the adjoint-kernel constraints governing second-order integrability [3, 4]. Arms treated the coupled Einstein–Maxwell problem on the appropriate electromagnetic bundle [5]. The Arms–Marsden–Moncrief momentum-map normal form then described the local solution space near a symmetric point as a smooth factor times the zero set of a homogeneous quadratic map [6, 8]. Higher-curvature Taub constraints and their second-order interpretation have also been developed in general form [9, 10].

Accordingly, this paper does *not* claim to discover the general principle "tangent cone equals moment-map zero." Its contribution is an explicit exhaustive realization in the declared fourth-order Weyl–Maxwell model: the complete certified finite carrier, a proof that exactly five covectors exhaust the exponential-polynomial cokernel, and a bounded refinement by independent polynomial-growth and characteristic-shell obstructions. The latter belongs to the broader problem of secular perturbation theory [11], but the concrete $\{\mu, \mathcal{P}, \mathcal{R}\}$ ledger and exceptional mixed resonance are model-specific results.

The relation is conceptual rather than a claim that the classical functional-analytic normal-form theorem has been reproved for this theory. Those theorems use Sobolev-completed initial-data manifolds, a constraint map with suitable splitting/Fredholm properties, gauge slices, and control of the full infinite-dimensional adjoint kernel. Here we instead prove an algebraic surjectivity and cokernel theorem on a complete *declared finite harmonic module* of the fourth-order Weyl–Maxwell equations. We do not establish the hypotheses needed to promote it to an infinite-mode Sobolev normal form. Conversely, the classical structural theorem does not by itself supply our block inventory, five-covector sufficiency statement, or bounded $\mathcal{P}/\mathcal{R}$ refinement.

3 Theory and common background

We use the compactified Plebański–Hacyan product

$$M = \mathbb{R}_t \times S^1_L \times S^2, \tag{4}$$

with unit sphere curvature and a parallel unit magnetic flux on a fixed compact $U(1)$ bundle. With the conventions of the companion linear phase-space paper, the background solves both Einstein–Maxwell and Weyl–Maxwell theory. The latter has local gauge group

$$\mathrm{Diff}(M) \times \mathrm{Weyl}(M) \times U(1).$$

The spatial slice $\Sigma = S^1 \times S^2$ is compact. Its background stabilizer algebra contains

$$H = \partial_t, \qquad P_x = \partial_x, \qquad J_1, J_2, J_3 \in \mathfrak{so}(3).$$

For a linear tangent u, the corresponding covariant phase-space moment maps are quadratic forms

$$\mu_X(u) = \frac{1}{2}\Omega(u, \mathcal{L}_X u), \qquad X \in \{H, P_x, J_1, J_2, J_3\}.$$

On a closed slice, the covariant Taub bridge identifies these with the pairings of the quadratic field-equation source against the stabilizer cokernel covectors.

Proposition 3.1 (Taub moment maps on ordinary and generalized modes)**.** *The five displayed automorphisms lift to the fixed magnetic bundle and act symplectically on every certified linear block. For every linear solution, including the homogeneous and twist Jordan blocks,*

$$\left\langle \zeta_X, \mathcal{E}^{(2)}_{\bar{\Phi}}(u,u) \right\rangle_\Sigma = \mu_X(u) = \frac{1}{2}\Omega_\Sigma(u, \mathcal{L}_X u), \qquad X \in \{H, P_x, J_1, J_2, J_3\}. \tag{5}$$

The right side is independent of the closed Cauchy slice Σ. There is no sixth $U(1)$ Taub covector in the fixed-bundle problem.

Proof. Twice differentiating the action Noether identity pairs the quadratic constraint source with the adjoint reducibility parameter and gives (5). Because u and $\mathcal{L}_X u$ solve the linearized equations, their Lee–Wald current is conserved. The difference between its integrals on two closed Cauchy slices is therefore the integral of an exact spacetime divergence and vanishes. This argument uses neither a frequency eigenbasis nor boundedness, so it applies to polynomial Jordan solutions.

Concretely, time translation acts on the homogeneous coordinates by

$$(a, b, c, d, Q_e, W_x) \longmapsto (b, 0, d, 2a, 0, Q_e)$$

and on each twist pair by $(A, B) \mapsto (B, 0)$. Substitution into the conserved Cauchy form gives, after omission of the common positive harmonic normalizations,

$$\mu_H^{\ell=0} = -a^2 - b^2 + bd - Q_e^2, \tag{6}$$

$$\mu_H^{\text{twist}} = 2|B|^2, \qquad \mu_{\mathbf{J}}^{\text{twist}} = -4A \times B. \tag{7}$$

Although the representatives contain powers of t, these expressions are constant; the apparent time-dependent contributions cancel in the conserved current.

The rotations lift as bundle automorphisms $\widehat{J}_a = (J_a, \chi_a)$, with $\iota_{J_a}\bar{F} + \mathrm{d}\chi_a = 0$ patchwise. A constant $U(1)$ parameter acts trivially on connection differences and is a reducibility parameter, not an independent Hamiltonian stabilizer. Finally, fixing the Chern class of the magnetic bundle forbids a harmonic magnetic variation. These facts leave exactly the five covectors in (5). $\square$

4 The complete finite harmonic carrier

The real finite-support carrier V_{fin} is the direct sum of the following certified linear blocks:

1. generic Einstein and extra primaries for every $\ell \geq 2$, parity, magnetic label, and compact momentum;
2. standard and extra exceptional dipoles for every compact momentum;
3. the axial twist Jordan block;
4. the six-coordinate homogeneous generalized block; and
5. the electric and Wilson global directions on the fixed bundle.

Reality pairs opposite frequencies and momenta. A finite input therefore produces a finite quadratic source. Its complete output decomposition is indexed by

$$(L, M, K, \Omega, \text{parity}).$$

Angular Clebsch–Gordan selection gives finitely many L, while temporal and circle products give finite sums of input frequencies and momenta.

Definition 4.1 (Finite correction modules)**.** Let $\mathscr{X}_{\text{ep}}$ be the spatially periodic fields whose time coefficients are finite sums of $t^p e^{-i\Omega t}$, with $p \in \mathbb{N}_0$ and a finite set of real frequencies. Let $\mathscr{X}_{\text{qp}} \subset \mathscr{X}_{\text{ep}}$ be the finite bounded quasiperiodic subclass with $p = 0$. Define

$$\mathcal{Z}_2^{\mathscr{X}} = \{u \in V_{\text{fin}} : \exists v \in \mathscr{X},\ \mathcal{L}v = -\mathcal{E}^{(2)}(u, u)\}.$$

Every element of $\mathscr{X}_{\text{ep}}$ is smooth, but $\mathscr{X}_{\text{ep}}$ is not the full C^∞, Sobolev, distributional, or retarded function space. The subscripts “ep” and “qp” are therefore load-bearing. A secular coefficient is finite at every finite time but prevents a uniformly bounded perturbative expansion. It may signal a nearby family with shifted frequency, a required renormalization of parameters, or genuine resonant growth. The certificates decide only these two finite-module dispositions.

5 Finite exponential-polynomial tangent-cone theorem

The sufficiency statement rests on a simple algebraic fact which we state explicitly because it is the step that distinguishes $\mathscr{X}_{\mathrm{ep}}$ from $\mathscr{X}_{\mathrm{qp}}$.

Lemma 5.1 (Exponential-polynomial surjectivity)**.** *Let $P(D_t) \neq 0$ be a constant-coefficient scalar differential operator. For every polynomial $f(t)$ and real Ω, there is a polynomial $g(t)$ such that*

$$P(D_t)\big(g(t)e^{-i\Omega t}\big) = f(t)e^{-i\Omega t}. \tag{8}$$

If $-i\Omega$ is a root of P of multiplicity m, one may take $\deg g \leq \deg f + m$. Consequently every nonzero invariant factor of a matrix operator over the fibre ring $K[D_t]$ is surjective on $\mathscr{X}_{\mathrm{ep}}$; only zero Smith factors contribute to its $\mathscr{X}_{\mathrm{ep}}$-cokernel.

Proof. Conjugation by $e^{-i\Omega t}$ replaces $P(D_t)$ by $P(D_t - i\Omega)$. Factor

$$P(z - i\Omega) = z^m Q(z), \qquad Q(0) \neq 0.$$

The operator D_t^m is surjective on polynomials, increasing the required degree by at most m. On each finite-dimensional polynomial-degree filtration, $Q(D_t) = Q(0)(1 + N)$ with N nilpotent, hence has the finite inverse $Q(0)^{-1}\sum_r(-N)^r$. For a matrix operator, unimodular Smith transformations preserve $\mathscr{X}_{\mathrm{ep}}$ and reduce the assertion to its scalar invariant factors. ☐

The complete output ledger after Noether restriction and local Diff $\times$Weyl $\times\, U(1)$ reduction is as follows. Here p and q are respectively the extra and Einstein characteristic polynomials; "zero factors" means persistent factors in the $\mathscr{X}_{\mathrm{ep}}$ cokernel, not roots of a nonzero characteristic polynomial.

Output stratum	Reduced operator or invariant factors	Zero factors	Cokernel interpretation
Generic $L \geq 2$, nonstatic	$1, 1, p, pq$; off-shell inverse and Lemma 5.1 on p/q shells	0	none
Generic $L \geq 2$, static	$p = -(3\Lambda + 3\kappa^2 - 2)/3 < 0$ and $q > 0$ for $\Lambda \geq 6$	0	none
Exceptional $L = 1$, nonstatic	invertible off shell; nonzero factors on $\Omega^2 - K^2 = 4, 4/3$	0	none in $\mathscr{X}_{\mathrm{ep}}$
Scalar $L = 0$, $(\Omega, K) \neq (0, 0)$	exact modulo local gauge and Noether identities	0	none
Homogeneous $L = 0$, $(\Omega, K) = (0, 0)$	constraints plus $D_t^4(C - K)$ and $D_t^2 A_x$	2	ζ_H, ζ_{P_x}
Axial dipole $L = 1$, $(\Omega, K) = (0, 0)$	twist polynomial operator; polar block invertible	3	$\zeta_{J_1}, \zeta_{J_2}, \zeta_{J_3}$
Remaining certified global/twist outputs	contained in the preceding $L = 0, 1$ blocks	0 additional	none additional

For completeness, put $\kappa := K_{\mathrm{out}}$ for the output circle momentum (unrelated to the homogeneous function $K(t)$ below). The static generic entries use

$$p = -\frac{3\Lambda + 3\kappa^2 - 2}{3} < 0, \qquad q = \Lambda^2 + 2\Lambda\kappa^2 - 2\Lambda + \kappa^4.$$

Putting $s = \Lambda - 6 \geq 0$ and $u = \kappa^2 \geq 0$ gives

$$q = s^2 + 2su + u^2 + 10s + 12u + 24 > 0,$$

so no static $L \geq 2$ zero factor is hidden in the table.

For the homogeneous source, quadratic products have degree at most six. The two dynamical invariants therefore have the explicit polynomial right inverses

$$D_t^4\left(\frac{t^{r+4}}{(r+1)(r+2)(r+3)(r+4)}\right) = t^r, \qquad D_t^2\left(\frac{t^{r+2}}{(r+1)(r+2)}\right) = t^r, \quad 0 \leq r \leq 6.$$

This displays rather than merely asserts the zero-block surjectivity.

Theorem 5.2 (Complete finite-support exponential-polynomial cone)**.** *For the complete carrier* V_{fin}*,*

$$\boxed{\mathcal{Z}_2^{\mathrm{ep}} = \{u \in V_{\mathrm{fin}} : \mu_H(u) = \mu_{P_x}(u) = \mu_{J_1}(u) = \mu_{J_2}(u) = \mu_{J_3}(u) = 0\}.} \tag{9}$$

Every point of this cone admits a real, spatially periodic second-order correction in $\mathscr{X}_{\mathrm{ep}}$*, and each of the five equations is necessary.*

Proof. Finite input support and Clebsch–Gordan selection produce finitely many output labels $j = (L, M, K, \Omega, \text{parity})$, with temporal polynomial degree at most six. The table exhausts those labels. Lemma 5.1 removes every nonzero invariant factor, including every nonzero-frequency resonance. The displayed polynomial primitives remove the homogeneous dynamical rows. Exactly five zero factors remain, so

$$\operatorname{coker} \mathcal{L}_{\mathrm{ep}} = \operatorname{span}\{\zeta_H, \zeta_{P_x}, \zeta_{J_1}, \zeta_{J_2}, \zeta_{J_3}\}.$$

Pairing the second-order equation with them and applying Proposition 3.1 proves necessity. Conversely, when all five moment maps vanish, every output source lies in the blockwise image. The finite corrections and their conjugates assemble a real element of $\mathscr{X}_{\mathrm{ep}}$, proving sufficiency. □

Remark 5.3. The theorem covers arbitrary finite sums across the certified generic, exceptional, homogeneous, and twist blocks. It is not a theorem for the full smooth, Sobolev, distributional, infinite-mode, or causal function space.

6 Complete bounded-obstruction decomposition

On $\mathscr{X}_{\mathrm{qp}}$, differentiation cannot produce a positive power of t, and a resonant pure exponential has no bounded quasiperiodic primitive. For each finite output block j, let $\mathcal{P}_{j,r}$ extract the coefficient of $t^r e^{-i\Omega_j t}$ for $r > 0$. After all such coefficients vanish, let $\mathcal{R}_{j,a}$ pair the remaining pure exponential with an exact basis of the reduced left kernel of $\mathcal{L}_j(-i\Omega_j)$. This definition includes the dynamical reduced left kernel at $\Omega_j = 0$. The stabilizer covectors inside the genuine zero block are split off and recorded separately as μ_X; no other zero-root covector is silently identified with them.

Theorem 6.1 (Complete finite bounded-obstruction ledger)**.** *For every* $u \in V_{\mathrm{fin}}$*, a second-order correction in* $\mathscr{X}_{\mathrm{qp}}$ *exists if and only if*

$$\mu_H(u) = \mu_{P_x}(u) = \mu_{J_i}(u) = 0, \qquad \mathcal{P}_{j,r}(u) = 0, \qquad \mathcal{R}_{j,a}(u) = 0 \tag{10}$$

for all finite output blocks and all corresponding indices. These are the complete linear obstruction types in the declared finite module. The common quadratic zero set of the resulting functionals is only partly classified.

Proof. If a bounded quasiperiodic v exists, then $\mathcal{L}v$ is also bounded and quasiperiodic, so every positive polynomial coefficient $\mathcal{P}_{j,r}$ vanishes. For each remaining pure-frequency block, finite-dimensional linear algebra says that $\mathcal{L}_j(-i\Omega_j)v_j = S_j$ is solvable precisely when S_j annihilates its reduced left kernel. At the genuine zero block those pairings split into the five Taub moment maps and any remaining dynamical zero-root functionals $\mathcal{R}_{j,a}$; at nonzero shells they are the finite-frequency $\mathcal{R}_{j,a}$. Off-shell blocks are invertible. Conversely, these conditions put every finite block source in the corresponding image, and the finite collection of solutions and conjugates assembles an element of $\mathscr{X}_{\mathrm{qp}}$. □

The certificates contain a moment-map-zero fixture with a nonzero $\mathcal{P}_{j,2}$ and another with a nonzero $\mathcal{R}_{j,a}$. Thus both new families are independent of the five global charges, and $\mathcal{Z}_2^{\mathrm{qp}} \subsetneq \mathcal{Z}_2^{\mathrm{ep}}$.

For finite generic oscillatory inputs the complete bounded $K = \Omega = 0$ receiver can be made fully explicit. Its nonzero source map is

$$(\mu_H, \mu_{P_x}, \mu_{J_1}, \mu_{J_2}, \mu_{J_3}, R_c), \qquad R_c = \frac{1}{2}\sum_j k_j^2 h_j.$$

Here h_j is the complete normalized Hermitian current block, including cross terms within a degenerate shell. The second formal homogeneous mean covector is the Wilson-acceleration row; its quadratic source vanishes by the integrated Maxwell identity. The polar $L = 1$ zero block contributes no physical cokernel, the axial block contributes exactly the three lifted rotations, and every static $L \geq 2$ block is reduced-invertible. Thus the six displayed conditions are necessary and sufficient for bounded inversion of the zero-frequency source. On the candidate-13 two-fibre carrier, joining them to the eighteen finite-frequency coefficients gives the complete formula

$$\mathcal{Z}_{2,13}^{\mathrm{qp}} = \{\mu_H = \mu_{P_x} = \mu_{J_1} = \mu_{J_2} = \mu_{J_3} = R_c = \mathcal{R}_{13,1} = \cdots = \mathcal{R}_{13,18} = 0\}.$$

This formula does not by itself certify a nonzero real point of that bounded zero locus. The explicit three-occupation fixture has $R_c < 0$, so it belongs to the smooth exponential-polynomial cone but not the bounded cone.

7 Complete standard generalized-zero subcone

The coordinates in the next theorem are concrete perturbations, not abstract labels. In coordinates (t, x, θ, ϕ), write

$$h_{xx} = C(t), \qquad h_{\theta\theta} = K(t), \qquad h_{\phi\phi} = K(t)\sin^2\theta, \qquad a_x = W_x + Q_e t, \tag{11}$$

$$K(t) = a + bt, \qquad C(t) = at^2 + \frac{b}{3}t^3 + c + dt. \tag{12}$$

For an orthonormal real $\ell = 1$ basis (Y_{1a}, X_a), the twist block is

$$h_{xA} = (A_a + B_a t)X_A^a, \qquad a_x = -(A_a + B_a t)Y_{1a}. \tag{13}$$

Coordinate	Linear representative	Time behaviour	Geometric meaning
a	sphere scale $K = a$ with induced $C = at^2$	constant/quadratic	homogeneous radion/Jordan seed
b	$K = bt$, $C = bt^3/3$	linear/cubic	homogeneous Jordan partner
c	$C = c$	constant	circle circumference modulus
d	$C = dt$	linear	circumference velocity
Q_e	$a_x = Q_e t$	linear	uniform electric tangent
W_x	$a_x = W_x$, $\mathrm{d}a = 0$	constant	flat circle Wilson line
$A \in \mathbb{R}^3$	twist h_{xA}, a_x	constant	lifted $SO(3)$ mapping-torus holonomy
$B \in \mathbb{R}^3$	same twist carrier	linear	twist velocity/Jordan partner

Theorem 7.1 (Standard generalized-zero bounded cone)**.** *On the complete standard generalized-zero carrier with homogeneous coordinates (a, b, c, d, Q_e, W_x) and axial twist position/velocity (A, B), bounded second-order extension holds exactly on*

$$a = b = Q_e = 0, \qquad B = 0, \qquad c, d, W_x \text{ arbitrary}, \qquad A \in \mathbb{R}^3 \text{ arbitrary}. \tag{14}$$

The surviving source has an explicit bounded correction.

The triangular positive-degree ideal supplies the proof. The complete homogeneous positive-degree coefficients are

Equation row	coefficient of t	coefficient of t^2
E_{11}	$15ab$	$15b^2/2$
sphere trace	$-15ab/2$	$-15b^2/4$
Maxwell x row	$Q_e a$	$Q_e b/2$

The remaining leading polynomial tensor is $\mathrm{STF}(B \otimes B)t^2$, whose squared real coefficient is $2|B|^4/3$. They force $b = B = 0$ and $Q_e a = 0$. Intersecting with the five moment maps leaves

$$\mu_H = -(a^2 + Q_e^2), \qquad \mu_{P_x} = 0, \qquad \mu_{\mathbf{J}} = 0,$$

and hence forces $a = Q_e = 0$ on this carrier. The remaining homogeneous source vanishes, while constant twist position is tangent to an exact lifted mapping-torus family and has a constant polar correction.

Explicitly, on the universal cover identify

$$(x, p) \sim (x + L, \exp(A^a J_a)p), \qquad p \in S^2, \tag{15}$$

and lift the rotation to the magnetic bundle with its patchwise Maxwell compensator. The local product fields descend by covariance. Differentiating this exact quotient family at $A = 0$ gives the constant twist representative; the stored quadratic correction is its constant polar $L = 2$ Taylor term.

Corollary 7.2 (Universal polynomial elimination)**.** *For every point of the complete finite-support bounded cone,*

$$b = 0, \qquad B = 0, \qquad Q_e a = 0.$$

Oscillatory products cannot cancel these zero-frequency polynomial rows.

8 Natural transport columns

Several apparently dangerous global–oscillator products are derivatives of exact families and can therefore be classified without solving each harmonic source independently.

Proposition 8.1 (Electric and Wilson columns)**.** *For every certified nonzero-frequency standard or extra primary, the electric-charge cross source has a bounded correction obtained by differentiating electromagnetic duality,*

$$F_\theta = \cos\theta F + \sin\theta \star_g F.$$

At the background,

$$h_{\mathrm{cross}} = 0, \qquad f_{\mathrm{cross}} = \star_{\bar{g}} f + (D_g \star)[h] F_{\mathrm{bg}}.$$

This is the mixed derivative at the purely magnetic background, not a claim that the finite duality orbit remains in one magnetic bundle component. Its sphere period vanishes for $\ell \geq 1$, so this mixed correction itself lifts globally as a fixed-bundle connection difference. Every Wilson–oscillator source vanishes because $\delta A = W_x dx$ has $\delta F = 0$ and the local action depends on the connection through F.

This removes the complete Q_e and W_x oscillator columns from the bounded frontier. It does not remove the separate homogeneous condition $Q_e a = 0$.

Theorem 8.2 (Complete circumference column)**.** *Let c vary the circle radius and let a certified oscillator have compact momentum k. The c-times-oscillator source admits a bounded correction if and only if*

$$c = 0 \qquad \text{or} \qquad k = 0.$$

For $k \neq 0$, the radius family gives a correction in $\mathscr{X}_{\mathrm{ep}}$ but none in $\mathscr{X}_{\mathrm{qp}}$.

Indeed, writing $R^2 = 1 + \eta c$,

$$\omega_R^2 = \frac{k^2}{R^2} + m_{\mathrm{branch}}^2, \qquad \partial_\eta \omega_R\big|_0 = -\frac{ck^2}{2\omega}.$$

Differentiating the exact family produces

$$\partial_\eta e^{-i\omega_R t}\big|_0 = \frac{ick^2}{2\omega} t e^{-i\omega t}.$$

To make the shell inference explicit, write the reduced matrix equation as $L(\omega_R, R)u_R = 0$, choose a left on-shell vector $u^\dagger$, and differentiate at $R = 1$. The Feynman–Hellmann identity is

$$\left\langle u^\dagger, (\partial_\eta L)u \right\rangle = -\omega_0' \left\langle u^\dagger, (\partial_\omega L)u \right\rangle = \frac{ck^2}{2\omega} \left\langle u^\dagger, (\partial_\omega L)u \right\rangle. \tag{16}$$

The last factor is the current-normalized shell form. It is nonzero on each standard branch and nondegenerate on every extra multiplicity block. Hence the particular circumference source, not merely some source, has nonzero adjoint pairing whenever $ck \neq 0$ and the corresponding mode coefficient is nonzero. This is a genuine $\mathcal{R}_{j,a}$ obstruction, not a polynomial $\mathcal{P}_{j,r}$ obstruction. At $k = 0$, $\omega_0' = 0$ and differentiating the covariant circle indices gives an ordinary same-frequency correction.

9 Exceptional resonance as an independent obstruction

Moment-map cancellation does not guarantee bounded or even nonsecular second-order closure in a restricted exceptional carrier. For the axisymmetric exceptional $\ell = 1, k = 0$ axial and polar amplitudes (a_x, a_p), a twist velocity can cancel the time-translation moment map. The two parity self-sources can also be tuned to cancel their even $L = 2$ resonance. But the required interior point has $a_x a_p \neq 0$, and the mixed axial $L = 2$ source has exact nonzero adjoint pairing

$$-\frac{8\sqrt{3}}{9}.$$

The boundary axes retain their individual bounded resonant self-obstructions. Thus the only axisymmetric two-polarization exceptional configuration in this carrier admitting a correction in $\mathscr{X}_{\mathrm{qp}}$ is $a_x = a_p = 0$. Nonzero configurations on the moment-map cone still admit the secular $\mathscr{X}_{\mathrm{ep}}$ corrections guaranteed by Theorem 5.2. The example proves that bounded resonant cokernel equations contain information independent of global charges; it does not contradict the exponential-polynomial theorem.

10 A complete tuned nonzero-momentum all-primary cone

The preceding exceptional result is a no-go on a restricted carrier. The next fixture shows that a bounded mixed cone can instead survive after every primary at the same angular and compact-momentum fibre is admitted. Fix

$$\ell = 2, \qquad m_A = 0, \qquad k^2 = 2\sqrt{3} - \frac{7}{6}, \tag{17}$$

choosing the circle circumference so that $\pm k$ are allowed compact momenta. The three positive input frequencies are

$$\omega_-^2 = \frac{29}{6}, \qquad \omega_e^2 = 2\sqrt{3} + \frac{25}{6}, \qquad \omega_+^2 = \frac{29}{6} + 4\sqrt{3}, \qquad \omega_- < \omega_e < \omega_+. \tag{18}$$

Here q_- and q_+ denote the two Einstein primaries, while X denotes the complete extra-primary multiplicity. Include both momentum signs, both axial and polar q_- amplitudes, arbitrary normalized X and q_+ multiplicities, and one constant twist position. No nonaxisymmetric input is included.

Write $A_\pm, P_\pm$ for the axial and polar q_- coefficients at momenta $\pm k$. The exact 140-row collision census covers every output $L = 1, 2, 3, 4$, momenta $K = 0, \pm 2k$, and every zero, sum, and difference frequency of q_-, X, and q_+. Its unique characteristic hit is the polar extra shell

$$L = 4, \qquad K = 0, \qquad \Omega = 2\omega_-, \tag{19}$$

generated only by the $q_- q_-$ product. Its two independent adjoint pairings are nonzero multiples of

$$A_+ A_- - 3P_+ P_-, \qquad A_+ P_- - A_- P_+. \tag{20}$$

Their complete complex zero set consists of the two mixed sheets

$$A_\pm = \sigma\sqrt{3}\, P_\pm, \qquad \sigma \in \{+1, -1\}, \tag{21}$$

and the two one-sided planes $A_+ = P_+ = 0$ or $A_- = P_- = 0$.

On a mixed sheet define the nonnegative Einstein-minus occupations

$$N_\pm = -h_-(A_\pm e_A + P_\pm e_P, A_\pm e_A + P_\pm e_P), \tag{22}$$

and let $E_\pm$ and $B_\pm$ be the total nonnegative occupations in the extra and Einstein-plus multiplicities. The complete Hamiltonian and circle momentum equations are

$$\omega_e^2(E_+ + E_-) + \omega_+^2(B_+ + B_-) = \omega_-^2(N_+ + N_-), \tag{23}$$
$$\omega_e(E_+ - E_-) + \omega_+(B_+ - B_-) = \omega_-(N_+ - N_-). \tag{24}$$

Theorem 10.1 (Tuned axisymmetric all-primary bounded cone)**.** *In the carrier just declared, a bounded finite-quasiperiodic second-order correction exists precisely at the origin or on one of the two mixed sheets* (21) *with*

$$\frac{1-r_e}{1+r_e} \le \frac{|P_+|^2}{|P_-|^2} \le \frac{1+r_e}{1-r_e}, \qquad r_e := \frac{\omega_-}{\omega_e}, \tag{25}$$

and with nonnegative positive-primary occupations satisfying (23)*–*(24)*. Their phases and multiplicity factorizations are otherwise arbitrary. The declared constant twist coordinate remains free and bounded-removable.*

Proof. Equations (20) give the four components displayed above. On either one-sided plane, positivity and $r_e < 1$ force the negative occupation to vanish when $\mu_H = \mu_{P_x} = 0$; hence only the origin survives. On a mixed sheet, the raw axial and polar weights cancel. At fixed positive-branch energy, the largest possible compact momentum is supplied by the lower positive frequency ω_e, so (23)–(24) have a nonnegative solution if and only if

$$|N_+ - N_-| \le r_e(N_+ + N_-), \tag{26}$$

which is equivalent to (25). Sufficiency is explicit: set $B_+ = B_- = 0$ and choose

$$E_+ + E_- = r_e^2(N_+ + N_-), \qquad E_+ - E_- = r_e(N_+ - N_-). \tag{27}$$

The interval makes both $E_\pm$ nonnegative. The five stabilizer pairings and the unique nonzero-shell adjoint pairings then vanish. Every other entry in the exact collision ledger is off shell, while the nonzero-Fourier scalar complex and constant-twist cross column are boundedly removable. Theorem 6.1 therefore supplies a bounded correction. Necessity follows from the same complete obstruction ledger. □

Because $\omega_- < \omega_e < \omega_+$, the interval (25) strictly contains the interval obtained when only the Einstein-plus primary is allowed to balance the negative branch. Thus the extra primary creates no new resonance at this fixture but does enlarge the bounded nonlinear cone. This is a theorem for one tuned $\ell = 2, m_A = 0, |k|$ fibre, not for nonaxisymmetric modes, other harmonics or circumferences, or multiple absolute-momentum fibres.

11 All-angular-momentum promotion and the two-momentum workload

11.1 The tuned standard-branch cone for every $\ell \ge 2$

The standard Einstein-minus calculation admits a uniform angular-momentum promotion. For each integer $\ell \ge 2$, tune the compact momentum by

$$k^2 = \sqrt{2\ell(\ell+1)} - \frac{\ell}{2} - \frac{1}{6}. \tag{28}$$

Let $a_\pm$ and $p_\pm$ denote the axial and polar Einstein-minus amplitudes at momenta $\pm k$. The complete standard-branch collision census has one relevant opposite-momentum characteristic shell. Up to nonzero exact normalizations $R_\ell > 0$ and $X_\ell \neq 0$, its two adjoint pairings are

$$\mathcal{R}_{\text{polar}} = R_\ell \left(a_+ a_- - \frac{\ell(\ell+1)}{2} p_+ p_- \right), \tag{29}$$

$$\mathcal{R}_{\text{axial}} = X_\ell \left(a_+ p_- - a_- p_+ \right). \tag{30}$$

Their complex zero variety consists of two mixed sheets

$$a_\pm = \sigma \sqrt{\frac{\ell(\ell+1)}{2}}\, p_\pm, \qquad \sigma \in \{+1, -1\}, \tag{31}$$

and the two one-sided planes on which one momentum sign vanishes.

Theorem 11.1 (All-ℓ tuned axisymmetric standard cone)**.** *For every $\ell \geq 2$ at the tuning (28), the complete bounded quasiperiodic cone on the declared axisymmetric standard Einstein-minus/Einstein-plus carrier is the origin together with the two mixed sheets (31) satisfying*

$$\frac{1 - r_\ell}{1 + r_\ell} \leq \frac{|p_+|^2}{|p_-|^2} \leq \frac{1 + r_\ell}{1 - r_\ell}, \qquad r_\ell := \frac{\omega_-}{\omega_+}. \tag{32}$$

The one-sided resonance planes meet the Hamiltonian and circle-momentum balance equations only at the origin.

The proof is uniform in ℓ: the exact collision census excludes every other standard-branch shell, equations (29)–(30) give the full resonance ideal, and the two global charge equations reduce to the elementary occupation interval (32). This is a genuine all-ℓ theorem, but it is still separately tuned at each ℓ, axisymmetric, and confined to the standard branches. It does not join different angular momenta or different absolute compact momenta.

11.2 Two absolute momenta: a branch-basis obstruction census

The first two-absolute-momentum enlargement has closed all 164 declared branch-basis coefficients. The axisymmetric $L = 4$ calculation is complete: all 108 coefficients across axial–axial, polar–polar, and the two ordered cross-parity sectors are exact. The nonaxisymmetric $L = 3$ calculation contributes 44 exact coefficients and the $L = 1$ companion the remaining 12. Every declared individual branch-basis fixture in these two blocks has nonzero cokernel and a bounded obstruction.

This census by itself is only an exhaustive coefficient statement for the named basis columns: different columns may cancel after amplitudes are joined. The candidate-13 fibre is the first case where the arbitrary-amplitude ideal, the zero-frequency receiver, and the correction-class-dependent range have all been joined, giving the complete formula displayed after Theorem 6.1. The analogous common zero varieties at the other collision circumferences remain open.

12 Current bounded frontier

The electric, Wilson, and circumference columns are closed, and the first full-time velocity repair sharpens the a, d-frontier. Earlier direct d-source fixtures were evaluated at $t = 0$. They correctly proved that the constant shell-projection matrix is invertible, but they could not prove that the full source was bounded. No frozen paper theorem asserted the stronger conclusion; the intermediate "complete d-column" reading is withdrawn and replaced by the following full-time statement.

Locality gives

$$\mathcal{E}^{(2)}[dt,u](t) = t\,\mathcal{E}^{(2)}[c,u](t) + \mathcal{E}^{(2)}[dt,u](0).$$

For the two axial extra representatives and the first polar representative, the coefficient of t vanishes. For the second polar representative it is the nonzero eight-row vector

$$(-36, 0, 164, -8\sqrt{3}i, 0, -100, -24, -20).$$

Thus the isolated fixture has polynomial zero locus $dz_{\mathrm{polar},2} = 0$. The previously computed constant-shell adjoint matrix remains valid, but cannot by itself establish boundedness. Combining this restored term with the four direct radion-position columns gives the complete positive-degree cross ideal on the $\ell = 2, k = 0$ extra multiplet:

$$\langle az_{\mathrm{ax},1}, az_{\mathrm{ax},2}, az_{\mathrm{pol},1}, az_{\mathrm{pol},2}, dz_{\mathrm{pol},2}\rangle. \tag{33}$$

These five monomials are already a Gröbner basis in the declared amplitude order. Their zero locus has three faces:

1. all extra amplitudes zero, with a, d unrestricted by this cross ledger;
2. $a = 0$ and $z_{\mathrm{pol},2} = 0$, with d and the two axial plus first polar amplitudes free;
3. $a = d = 0$, with all four extra amplitudes free.

This is a complete $\mathcal{P}$-zero locus for the declared cross block, not by itself a complete bounded cone. Before the other conditions are imposed, its surviving faces must still satisfy moment maps, constant shell pairings, self-products, and twist cross terms. On the complete global-plus-extra carrier those remaining conditions now close as follows.

Theorem 12.1 (Complete global plus $\ell = 2, k = 0$ extra bounded cone)**.** *Adjoin all four axial/polar $\ell = 2, k = 0$ extra-primary amplitudes $(x_{\mathrm{ax},1}, x_{\mathrm{ax},2}, x_{\mathrm{pol},1}, x_{\mathrm{pol},2})$ to the complete homogeneous and twist carrier. The $\mathscr{X}_{\mathrm{qp}}$ tangent cone is exactly*

$$\mathcal{Z}^{\mathrm{qp}}_{2,\,\mathrm{global}+\ell 2X} = \{(c, d, W_x, A) : c, d, W_x \in \mathbb{R},\ A \in \mathbb{R}^3\}. \tag{34}$$

In particular, it contains no nonzero $\ell = 2, k = 0$ extra-primary direction.

Proof. The universal polynomial rows first force $b = B = 0$ and $Q_e a = 0$. The remaining Hamiltonian moment map is

$$\mu_H = -a^2 - Q_e^2 - \frac{4}{3}X, \tag{35}$$

$$X = 1296|x_{\mathrm{ax},1}|^2 + \frac{208}{3}|x_{\mathrm{ax},2}|^2 + 22464|x_{\mathrm{pol},1}|^2 + 12288|x_{\mathrm{pol},2}|^2. \tag{36}$$

It is a strictly negative sum of squares, so its real zero locus has $a = Q_e = x_{\mathrm{ax},1} = x_{\mathrm{ax},2} = x_{\mathrm{pol},1} = x_{\mathrm{pol},2} = 0$. What remains is precisely the standard cone of Theorem 7.1, whose bounded correction is already explicit. The repaired ideal (33) is automatically satisfied there. □

After these repairs, the still-open full-carrier work is:

1. standard Einstein oscillators and exceptional $\ell = 1$ inputs outside the separately classified fixtures, other harmonics, and generic nonzero momenta;

2. their a, d polynomial columns and shell-pairing zero loci;
3. shell resonances among bounded wave products and remaining mixed blocks; and
4. the common zero locus for arbitrary finite sums.

Any future completion must preserve the separation between positive-degree obstructions $\mathcal{P}$ and shell pairings $\mathcal{R}$.

Carrier or column	$\mathscr{X}_{\mathrm{ep}}$ correction	$\mathscr{X}_{\mathrm{qp}}$ correction
Complete finite-support cone	iff five moment maps vanish	additional $\mathcal{P}$ and $\mathcal{R}$ conditions
Standard generalized-zero block	classified	classified completely
Global plus $\ell = 2, k = 0$ extra	moment-map cone contains balanced extra directions	bounded cone equals (c, d, W_x, A); no extra wave
Electric charge $\times$ oscillator	finite quasiperiodic	removable by duality transport
Wilson $\times$ oscillator	zero source	zero source
Circumference $\times$ $k = 0$ mode	ordinary same-frequency	removable
Circumference $\times$ $k \neq 0$ mode	secular radius derivative	obstructed
Velocity $d\times$ selected polar extra mode	finite polynomial source	nonzero $\mathcal{P}_{j,1}$ unless canceled jointly
Exceptional axial–polar interior	secular primitive on moment cone	bounded-resonance obstructed
Tuned $\ell = 2, m_A = 0$, nonzero $\lvert k\rvert$, all primaries	moment-map cone	Theorem 10.1: two mixed sheets over the exact occupation polytope
$a, d \times \ell = 2, k = 0$ extra	allowed on moment cone	polynomial ideal (33), then $\mathcal{R}$ still required
Remaining a, d oscillator columns	allowed by ep theorem on moment cone	open

13 Interpretation

The solution set near a symmetric compact background is not generally a linear space. Its second-order approximation is a cone cut out first by a moment map and then, if uniform boundedness is demanded, by resonance conditions. Three points are important.

First, a linearly nonradical extra Weyl mode may fail to be tangent to the nonlinear fixed-charge solution space. This is nonlinear selection, not linear gauge removal. Second, mixing Einstein and extra directions can cancel the moment maps, so no conclusion follows from the sign of a pure branch alone. Third, even complete moment-map cancellation leaves bounded resonances. Exponential-polynomial formal continuation and bounded perturbative stability are therefore different claims.

The tuned nonzero-momentum theorem adds a complementary point: enlarging the positive branch can widen a surviving bounded cone without introducing a new shell collision. The extra primary is therefore neither uniformly deleted nor uniformly liberated by the quadratic test; its disposition depends on the complete carrier and correction class.

The result supplies the middle map in the programme's residual atlas:

$$X_\alpha \longmapsto \Omega(X_\alpha, \overline{X}_\alpha) \longmapsto \mu(X_\alpha) \longmapsto D^2\mathcal{E}[X_\alpha, X_\alpha].$$

Interaction sourcing, detector response, and BRST status are separate later maps.

14 What is not proved

The following remain outside the theorem:

- infinite harmonic sums and Sobolev completion;
- retarded or advanced solvability and nonlinear causal dependence;
- integration of a second-order jet to an exact or all-orders family;
- the final residual/background-stabilizer quotient;
- asymptotically flat, de Sitter, anti-de Sitter, or black-hole boundary conditions;
- observer coupling, particles, quantum BRST cohomology, and unitarity.

These exclusions are not cosmetic. In particular, Theorem 5.2 does not establish nonlinear stability.

15 Certificate and reproduction ledger

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Repaired $a, d \times \ell = 2$ polynomial zero locus	`bridge/certificates/einstein_maxwell_weyl_ad_ell2_extra_polynomial_zero_locus.json`
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All-ℓ standard-branch resonance and bounded cone	`bridge/certificates/einstein_maxwell_weyl_symbolic_ell_tuned_axisymmetric_bounded_cone.json`
Two-momentum axisymmetric $L = 4$ matrices	`bridge/certificates/einstein_maxwell_weyl_ell2_two_abs_momentum_axial_axial_L4_matrix.json` and the three parity companion matrices
Two-momentum nonaxisymmetric $L = 3$ matrix	`bridge/certificates/einstein_maxwell_weyl_ell2_two_abs_momentum_nonaxisymmetric_L3_matrix.json`
Two-momentum nonaxisymmetric $L = 1, 3$ completion	`bridge/certificates/einstein_maxwell_weyl_ell2_two_abs_momentum_nonaxisymmetric_L1_L3_matrix.json`
Moment-map/Taub bridge	`bridge/certificates/einstein_maxwell_weyl_moment_map_taub_bridge.json`

Each certificate records its schema, imported hashes, exact commands, test tier, assumptions, and missing-object ledger. The independent verifier for each named certificate must pass before this manuscript advances to THEOREM-FROZEN. The current draft promotes only the scoped complete formulas displayed above; the real component geometry of the unrestricted finite-support bounded zero set remains open.

16 Outlook

The immediate mathematical target is the real component geometry and nontrivial-point problem for the complete bounded finite-support functional formula. The symbolic-ℓ standard-branch resonance and collision theorem is now closed, as is the complete 164-coefficient two-momentum branch-basis census. The next exact enlargement is the common amplitude-level zero variety, followed by a join of distinct $|k|$ fibres without identifying their phase carriers. The immediate physical target is to compare its strata with the transferred Einstein/extra/Maxwell interaction tensor and with observer response. A later causal theorem would replace the finite secular right inverses by support-controlled Green operators or prove that this cannot be done.

The intended final synthesis is not that every fourth-order mode disappears. It is that each mode receives an exact disposition:

$$\boxed{\text{linear mode} \;\rightarrow\; \text{moment map} \;\rightarrow\; \text{resonance} \;\rightarrow\; \text{interaction and observation.}}$$

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